MILITARY SPECIFICATION

SHELTER, ELECTRICAL EQUIPMENT S-404( )/MSC

1. SCOPE

1.1 Scope.- This specification covers a transportable shelter facility which is part of and incorporates all facilities for the installation and operation of components of communicating Patch Panel SB-675( )/MSC for use within the Army Area Communications System.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-V-173	Varnish, Moisture and Fungus Resistant
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronic and Associated Electrical Equipment)
MIL-W-6858	Welding, Resistance
MIL-T-7928	Terminals, Lug and Splice, Crimp Style Copper
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-P-55505	Shelter, Electrical Equipment (containing equipment) Packaging and packing of

STANDARDS

MILITARY

MIL-STD-22	Welded - Joint designs
MIL-STD-105	Sampling Procedures and Tables for Inspection by attributes
MIL-STD-252	Wired Equipment, Classification of Mechanical and Visual Defects

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DRAWINGS

DL-SC-A-595474	Supplemental data list, Shelter, Electrical Equipment S-404( )/MSC
SC-E-595475	Signal Schematic Wiring Diagram S-404( )/MSC

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description.- Shelter, Electrical Equipment S-404( )/MSC (hereafter referred to as shelter facility) is a modified S-280( )/G shelter, which, upon the installation of certain operating equipment becomes as SB-675( )/MSC. The SB-675 ( )/MSC is a versatile mobile switching unit capable of terminating and interconnecting 936 local circuits. Test facilities are provided for testing all local circuits. One (1) TT-98( )/FG is provided for monitoring teletypewriter circuits. Cable connectors and binding posts are provided for making connections to other components in the area communications center. The shelter facility requires 115 VAC 60 cycle single phase primary power.

3.2 Preproduction sample.- Unless otherwise specified in the invitation for bids and contract, the contractor shall furnish one (1) preproduction sample for the S-404( )/MSC for approval. (See 4.3)

3.3 Operation equipment.- Where specified herein, the following government loaned equipment shall be installed in the shelter facility to be tested, unless otherwise specified in the bid request or contract.

1 each	Switchboard, Telephone Manual SB-22( )/PT	
1 each	Intercommunication Station LS-147C/FI	
2 each	Telephone Set TA-312( )/PT	
1 each	Terminal Telegraph TH-22( )/TG	
1 each	Test Set TS-140( )/PCM including:	1 each SG-150/PCM 1 each ME-22A/PCM
1 each	Multimeter ME-77( )/U	
1 each	Teletypewriter TT-98/FG	
1 each	Headset, Micorphone, H-144C/U	

3.4 Construction.-The shelter facility shall be constructed in accordance with the model supplied, the list of exceptions to the model, schematic SC-E-595475 and drawings of components as listed on DL-SC-A-595474.

3.5 Electrical installation.- Shelter facility S-404/MSC

3.5.1 Power wiring.- All AC power wiring shall be in accordance with the model and comply with the requirements of this specification.

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3.5.1.1 Dielectric breakdown and insulation resistance, power wiring.- All AC power wiring shall have an insulation resistance of not less than 100 megohms when measured between

a. "Hot line" and "Neutral"

b. "Neutral" and "Ground"

c. "Hot line" and "Ground"

Insulation resistance shall be measured no less than 10 seconds after a potential of 500 VDC is applied instantaneously, and with the voltage still applied. (See 4.6.1.9)

3.5.2 Signal wiring.-The signal wiring shall be in accordance with the schematic provided, and the model unless otherwise noted in the contract, and shall comply with the requirements of this specification.

3.5.2.1 Continuity, signal wiring.- There shall be electrical continuity between each terminal as shown on the drawings provided.

3.5.2.2 Cord assemblies.- Patch cords and air conditioner cable assemblies shall be constructed in accordance with MIL-C-3885.

3.5.2.3 Dielectric breakdown and insulation resistance, signal wiring.- All signal wiring including individual wires, twisted pairs, cable harnesses, and cable assemblies shall have an insulation resistance of not less than 100 megohms when measured between each conductor and the remaining conductors, metallic connector shells, conduit, wireduct, and shelter ground. Insulation resistance shall be measured at least 10 seconds after an instantaneous potential of 500 volts DC is applied (See 4.6.2.3).

3.6 Military truck transport.- The shelter facility shall be capable, without any extra tying, taping, etc., of being transported over cross country terrain by military vehicle and shall withstand the test of 4.7. After testing per 4.7, there shall be no deformation, buckling, delamination, or damage to any part of the shelter facility.

3.7 Rail impact test.- The shelter facility shall be capable, without any extra tying, taping, etc., of being loaded, blocked and braced on a flatcar for shipment by rail, and shall withstand the test of 4.8. After each test there shall be no deformation, buckling, delamination or damage to any part of the shelter facility.

3.8 Drops, flat and rotational.- The shelter facility shall withstand, without any extra tying, taping, etc., drops of 18 inches onto concrete. After each of the drop tests of 4.9.1 and 4.9.2 there shall be no deformation, buckling, delamination or damage to any part of the shelter facility.

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3.9 Watertightness.- The shelter facility shall withstand the test of 4.10. After the test, there shall be no leakage into the shelter facility or its walls, floor, roof, or doors.

3.10 Fording.- The shelter facility prepared for transportation on a military cargo truck, shall withstand the test of 4.11. After the test, there shall be no leakage into the shelter facility or its walls, floor, or doors.

3.11 Mounting inserts.- All mounting inserts shall be installed without the use of any auxiliary materials such as adhesive and when installed the head shall not protrude more than 0.062 inch from the inside skin of the shelter. When installed each insert shall be capable of withstanding the tensile load and torque load referenced in paragraph 4.13. There shall be no damage to the insert threads, the insert shall not have rotated from its original position, and there shall be no distortion of the panel, mounting member, or insert.

3.12 Cleaning.-

3.12.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in an applicable portion of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled.

3.12.2 Units.- After assembly, the equipment shall be cleaned thoroughly and shall be free from excess solder or flux, loose hardware, and other foreign material.

3.13 Finish, protective.- The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings (See 4.4).

3.14 Marking.-

3.14.1 General.- Marking shall conform to Specification MIL-M-13231.

3.14.2 Nameplates.- Each electrical equipment shelter facility S-404( )/MSC shall be serial numbered and provided with nameplates located as on the model.

3.14.3 Aircraft loading data plate.- The data plate for the S-280/G shelter shall be covered with the data plate of the shelter facility.

3.14.4 Interior and exterior marking.- Interior and exterior markings shall be as depicted in the model.

3.15 Riveting.- Riveted joints shall be tight, the joined points shall be undamaged and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.16 Soldering.-

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3.16.1 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections but no acid or acid salts shall be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above they shall be completely neutralized and removed immediately after use.

3.16.2 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heat during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.17 Tropicalisation of material.- Canvas and materials as indicated in the model shall be treated to be resistant to moisture and fungi as follows:

3.17.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.17.2 Toxicity.- Treating materials shall cause not skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished item when used for the purpose intended.

3.17.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements.

3.17.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use (See 6.1).

3.18 Welding.- Welded joints shall conform to Standard MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary. Finished welds shall not be ground. Spot, stitch, and seam welds shall conform to Specification MIL-W-6858, Class B. All surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from scale, paint, grease and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning

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shall be completely neutralized and removed (See 4.4).

3.19 Wiring and cabling.- Wiring and cabling shall be neat and sturdy.

3.19.1 Cabling.- Insulated wires shall be formed into cables as indicated on drawings. Individual conductors thus secured shall lie essentially parallel; however, this does not prohibit the use of twisted pairs.

3.19.2 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided for at least two replacements of the part to which the wire or cable is connected.

3.19.3 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.19.4 Splicing.- Wires shall not be spliced.

3.19.5 Connections.-

3.19.5.1 General.- Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver-plated, or lead-alloy coated. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to Specification MIL-V-173. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.19.5.2 Crimping of solderless terminal lugs.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of Specification MIL-T-7928 (See 4.12).

3.20 Seal, vapor.- All joints and raw edges of the framing for the blower exhaust, power and signal entrance boxes, and air conditioner cutouts shall be sealed to provide a vapor barrier against the entrance of moisture to the core material of the shelter panels and to the interior of the shelter. All joints and edges shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws, and fasteners shall be dipped in sealer prior to installation. The sealer shall have the properties of 3.20.1, 3.20 2, and 3.20 3 (See 4.14)

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3.20.1 Sealer shear at temperature.- The minimum shear strength of the fully cored sealer and its bond to the aluminum shall be as follows:

200 psi at ambient (80°F ± 15°)
200 psi at -65°F
75 psi at 200°F

3.20.2 Sealer shear after humidity exposure.- The sealer shall withstand exposure to 100 percent relative humidity at 160°F for 2 weeks and have a shear strength of 200 psi minimum.

3.20.3 Sealer shear after salt spray exposure.- After 2 weeks exposure to 20 percent Na Cl solution at 95°F the sealer shall have a shear strength of 200 psi minimum.

3.21 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts, such as mounts, brackets, cabinets, etc., shall be physically and functionally interchangeable, without modification of such items, or of the shelter, or of the mounting locations (See 4.15). Individual items shall not be hand-picked for performance or to fit the assigned position, mount or bracket. Reliance shall not be places on any unspecified dimension, rating, characteristic, etc.

3.22 Workmanship.- The equipment shall be manufactured and assembled in accordance with applicable portions of the following paragraphs:

3.12 Cleaning
3.13 Finish
3.14 Marking
3.15 Riveting
3.16 Soldering
3.17 Tropicalization of material
3.18 Welding
3.19 Wiring and Cabling
3.20 Seal, vapor

4. QUALITY ASSURANCE PROVISIONS.-

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be as classified as follows:

a. Preproduction inspection (does not include preparation for delivery) (See 4.5)

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b. Inspection covered by subsidiary documents (See 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (See 4.5).

(2) Quality conformance inspection of preparation for delivery (See 4.16).

4.3 Preproduction inspection.- This inspection shall be performed by the contractor on the First Article unless otherwise specified in the contract. It shall consist of the inspection specified in Table I, the inspection specified in the subsidiary documents covering the items listed in 4.4 and the inspection specified for group A, group B, and group C (See Tables II, III and IV respectively). The preproduction inspection shall be performed in this order: (a) Mounting inserts, (b) Power wiring, (c) Signal wiring, (d) Military truck, (e) Rail transport, (f) Flat drop, (g) Rotational drop, (h) Watertightness, (i) Fording. The last tests performed shall be a repeat of the electrical tests. The remaining preproduction tests shall be performed in any order satisfactory to the contracting officer.

Table I - Preproduction Inspection

Inspection (For additional preproduction inspection see 4.3)	Req't Para	Insp Para
Military truck transportation	3.6	4.7
Rail transportation	3.7	4.8
Sealer, shear	3.20	4.14

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

ITEM	WHERE REQUIRED
Finish	3.13
Marking	3.14
Welding	3.18

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 for 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review

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and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government - at its discretion - may perform all or any part of the specified inspections, to verify the contractor's compliance with specified requirements (See 6.5). Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to Table II and the inspection procedures of Standard MIL-STD-105, using the general inspection levels. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table II - Group A Inspection

Inspection	Req't Para	Insp Para	AQL Major	AQL Minor
Visual and Mechanical	3.22	4.16	1.5 dphu**	6.0 dphu*
Electrical				
Power wiring	3.5.1	4.6.1	1% for the entire group of power and signal wiring	
Operation of distribution panel circuits	3.5.1.1	4.6.1.1 thru 4.6.1.8		
Dielectric breakdown and insulation resistance, power wiring	3.5.1.2	4.6.1.9		
Signal wiring	3.5.2	4.6.2		
Distribution and contunuity	3.5.2.1	4.6.2.1 4.6.2.2		
Patch cords	3.5.2.2	4.6.2.3		
Dielectric breakdown	3.5.2.3	4.6.2.4		

*All electrical defects are considered major
**DPHU - defects per hundred units

4.5.2 Group B inspection.- This inspection, including sampling, shall conform to Table III and to the procedures for all small-sample inspections of Standard MIL-STD-105, using the special inspection levels. Group B inspection shall normally be

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performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.5.2.1 Group B sampling plans.- The group B sampling plans, for AQL's listed in Table II, shall be as follows:

AQL	Inspection level for normal and tightened and reduced inspection
4.0%	S-4
6.5%	S-4

4.5.2.2 Order of inspection within group B.- Group B inspection shall be performed in the order listed in Table III.

Table III - Group B Inspection

Inspection	Req't Para	Insp Para	AQL (see 4.5.2.1)
Insert test	3.11	4.13	4.1%
Watertightness	3.9	4.10	6.5% for group
Fording	3.10	4.11	
Interchangeability	3.21	4.15	4.0%

4.5.3 Group C inspection.- This inspection shall consist of the tests specified in Table IV and shall be performed on sample units that have been subjected to and met groups A and B inspection. Sample units shall be selected in accordance with paragraph 4.5.3.1.

Table IV - Group C Inspection

Inspection	Req't Para	Insp Para
Solderless-terminal-lug connection	3.19.5.2	4.12
Flat drop	3.8	4.9.1
Rotational drop	3.8	4.9.2

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4.5.3.1 Sampling for inspection.- One sample shall be selected at random for each group of inspections in Table IV from the first 50 units, or fraction, thereof, produced and one sample from each additional 150 units and fraction thereof produced. The first sample selected shall not be from the first items produced.

4.5.3.2 Sampling for inspection of solderless-terminal-lug connections.- Five specimens of each combination of wire and terminal shall be prepared each week for inspection. A specimen is a 1 foot length of wire on which a terminal lug has been applied using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.3 Noncompliance.- The contractor shall immediately report in writing each group C failure occurence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspection which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the group C reinspection results have shown that the corrective action was effective.

4.5.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed both group B and group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

4.6 Electrical installation tests.- The tests of 4.6.1 and 4.6.2 shall be performed on each shelter facility. Before applying power to the shelter facility, an electrical ground at the power entrance box shall be made. Power to be supplied shall be 115 VAC, 60 cycles, single phase.

4.6.1 Test of main breaker circuit.-

a. Remove neon power indicator lamp. Switch main breaker to off.

b. Using a low voltage indication device (such as a buzzer) verify continuity between terminals 1 of one Power Receptacle U-238( )/G and terminal 1 of other U-238( )/G in Power Entrance Box.

c. Repeat b for terminals 2 and 4 of both receptacles.

d. Install neon power indicator lamp.

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e. Connect a 115 VAC 60 cycle power source to the facility through one of the U-238( )/G power receptacles. The neon power indicator lamp shall be lighted, and the voltmeter shall indicate a nominal 115 VAC.

4.6.1.2 Test of breaker No. 1 circuit.-

a. Switch on breaker No. 1, neon indicator shall light.

b. Turn on "blackout bypass" switch.

c. Turn on "cold start" lamp switch, incandescent lamps shall light.

d. Turn on light switch No. 1. The two inboard rows of fluorescent lamps shall light.

e. Turn on light switch No. 2. The two outboard rows of fluorescent lamps shall light.

f. Close door and turn off blackout bypass switch. All lamps shall remain on.

g. Unlatch door. All lamps shall entinguish before latch is in a full open position. Neon power indicator lamp shall remain on.

h. Relatch door. All lamps shall be on when latch is in the locked position.

i. Repeat f and g twice.

j. Turn on blackout bypass switch and leave on for remainder of inspection. All lamps shall be on.

k. Switch breaker No. 1 off and then on. All lamps shall alternately extinguish and relight.

4.6.1.3 Test of breaker No. 2 circuit.-

a. Switch on breaker No. 2, neon indicator shall light. (Check that vent cover is open). Plug in blower.

b. Turn on switch for blower. Blower shall operate and shall exhaust air from shelter.

c. Turn off switch for blower. Blower shall stop.

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4.6.1.4 Test of breaker No. 3 circuit.-

a. Switch on breaker No. 3, neon indicator shall light. Plug electric heater No. 1 into heater No. 1 receptacle.

b. Set switch on heater to HEAT position and turn thermostat to temperature higher than ambient. Heater shall blow warm air after a few minutes of operation.

c. Turn thermostat down to a temperature colder than ambient. Heating coil shall turn off indicated by a drop in the ammeter reading. Fan shall continue to blow air.

d. Set switch to a straight up position. Heater shall turn off.

e. Set switch to FAN position. Heater shall blow air.

f. Switch off breaker No. 3. Heater shall turn off.

4.6.1.5 Test of breaker No. 4 circuit.- Repeat test in paragraph 4.6.1.4 for breaker No. 4 and heater No. 2.

4.6.1.6 Test of breaker No. 5 circuit.-

a. Switch on breaker No. 5, neon indicator shall light.

b. Plug 115 VAC test lamp into each side of all "Convenience" duplex receptacles. Test lamp shall light in each position.

c. Switch off breaker No. 5. The test lamp shall not light when plugged into any of the "Convenience" receptacles.

4.6.1.7 Test of breaker No. 6 circuit.-

a. Switch on breaker No. 6, neon indicator shall light.

b. Plug 115 VAC test lamp into each equipment receptacle.

c. Switch off breaker No. 6. The test lamp shall not light when plugged into any equipment receptacle.

4.6.1.8 Test of breaker No. 7 circuit.-

a. Switch on breaker No. 7, neon indicator shall light.

b. Connect air conditioner power cable to receptacles on air conditioner and shelter.

c. Turn air conditioner fresh air switch to OPEN position and fan-cool switch to FAN. Air current shall emit from air conditioner.

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d. Turn fan-cool switch to COOL position and fresh air switch to CLOSED position. Turn thermostat control to lowest temperature. After a period of about 15 minutes the temperature of the air from the air-conditioner shall be cooler than the ambient air in the shelter. The unit shall function smoothly and the ammeter reading shall have increased above the first observed reading.

e. Turn off breaker No. 7, air-conditioner shall stop functioning. Return fan-cool switch to OFF position.

4.6.1.9 Dielectric breakdown and insulation resistance, power wiring.- The dielectric breakdown and insulation resistance of the power wiring shall be tested for conformance with 3.5.1.2 as described below.

a. Remove all fluorescent lamps from their fixtures.

b. Remove all starters from fluorescent fixtures.

c. Remove cover from power distribution box. Disconnect voltmeter from circuit. Disconnect all green (ground) wires from neutral bus bar in power distribution box. Switch on all circuit breakers and switches.

d. Remove all incandescent and neon lamps.

e. Unplug all power cords (blowers, air conditioners, heaters, intercom, radio, power supply etc.) from receptacles in wireduct.

f. Apply an instantaneous voltage of 500 volts DC between terminal 1 of Power Receptacle U-238( )/G and terminal 2 of the same receptacle. After 10 seconds measure the resistance between terminal 1 and terminal 2. Resistance shall not be less than 100 megohms.

CAUTION. AFTER EACH TEST, DISCHARGE THE CONDUCTOR UNDER TEST TO GROUND TO ELIMINATE DANGER OF SHOCK.

g. Repeat f replacing terminal 2 (2 places) with terminal 4.

h. Repeat f replacing terminal 1 (2 places) with terminal 4.

i. Restore shelter facility to original condition; that is, replace all lamps, cover, wires, etc.

4.6.2 Signal wiring, facility.- The signal wiring installation shall be inspected for conformance with the model. Jack designations (such as A16J24 etc.) are as follows:

a. The first letter defines which panel the particular jack is located. A is the panel on the left, B is the center panel, and C is the panel on the right. (With the viewer facing the jack panels.)

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b. The digit or digits following is the number of the jack strip, numbering consecutively from the top, starting with (1).

c. The digit or digits following the (J) indicate the jack number, with the numbering from the left to the right, starting with (1).

4.6.2.1 Continuity test fixtures.- Test fixtures shall be fabricated in the following manner, and shall hereafter be identified by the referenced title.

a. Fixture No. 1 jumper together all terminals of a J-1077 junction box. Connect one end of CX4556/C cable to box.

b. Fixture No. 2 same as a above.

c. Fixture No. 3 connect 26 phone plugs to bare end of 15 foot stub signal cable. Solder tips of plugs to "A" terminals, and rings to "B" terminals. (The plugs should be numbered from 1 to 26 and connected to the corresponding pins of the connector. Mate connector end of cable to J-1077 box prepared as in a above).

d. Fixture No. 4 same as c above.

e. Fixture No. 5 stub signal cable with no terminations on bare end other than tinned leads, connector end terminated in junction box as in a above.

f. Shorted plug. Connect a diode across the tip and ring of a phone plug, conducting from tip to ring.

g. Reverse shorted plug. Connect a diode across the tip and ring of a phone plug conducting from ring to tip.

h. Dummy plug. A phone plug with discontinuity between tip and ring.

i. Continuity tester. A low voltage continuity indicating device, in series with a battery power supply, connected to a phone plug. The positive side of the unit shall be connected to the tip of the phone plug, and the negative side to the ring.

j. Fixture No. 6. Place jumper wires on a J-1077 junction box paralleling all the "A" pins. Repeat for all "B" pins. Mate connector of CX4566/C cable to one connector of box. Mate connector of another CX4566/C cable to other connector of box.

k. Resistance phone plug. A phone plug terminating in an ohmeter. (NOTE: The ME77/U multimeter may be used, by patching in the correct jacks.)

l. Shorted connector. A mate to the U-187/G connector, with all pins shorted together (36 each will be required).

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m. Fixture No. 7. Same as c above.

4.6.2.2 Continuity tests.-

a. Switched pairs.

(1) With continuity tester, verify continuity between pin 26A of connector #1 and tip of intercom jack in wiremold.

(2) Connect one end of fixture No. 6 to connector #1 of shelter.

(3) Connect the other end of fixture No. 6 to connector #2 of shelter.

(4) Insert continuity tester into intercom jack in wiremold. The test shall indicate open circuit.

(5) Insert shorting plug into all jacks on jack strips A1 through A4 in sequence. Continuity shall be indicated for all insertions.

(6) Repeat steps (3) through (5) for -

(a) Connectors 3 and 4 for jack strips A5 through A8.

(b) Connectors 13 and 14 for jack strips B1 through B4.

(c) Connectors 15 and 16 for jack strips B5 through B8.

(d) Connectors 25 and 26 for jack strips C1 through C4.

(e) Connectors 27 and 28 for jack strips C5 through C8.

(f) Connectors 5 and 6 for jack strips A9 through A11.

(g) Connectors 7 and 8 for jack strips A12 through A14.

(h) Connectors 9 and 10 for jack strips A15 through A17.

(i) Connectors 11 and 12 for jack strips A18 through A20.

(j) Connectors 17 and 18 for jack strips B9 through B11.

(k) Connectors 19 and 20 for jack strips B12 through B14.

(l) Connectors 21 and 22 for jack strips B15 through B17.

(m) Connectors 29 and 30 for jack strips C9 through C11.

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(n) Connectors 31 and 32 for jack strips C12 through C14.

(o) Connectors 33 and 34 for jack strips C15 through C17.

(p) Connectors 35 and 36 for jack strips C18 through C20.

(q) Connectors 23 and 24 for jack strips B18 and B19.

b. Direct continuity.

(1) Remove all telephone, teletype, patch cord, and intercom plugs from jacks in wireduct and patch panel. Check that audio lines are disconnected from SB-22 and TH-22's, and are not shorted to each other, the shelter, or any other metallic object.

(2) Insert shorted plug into A2J1.

(3) Insert continuity tester into A3J1. The test shall indicate continuity.

(4) Insert dummy plug into A1J1. Test shall indicate open circuit. Remove plug and continuity shall be indicated.

(5) Insert dummy plug into A4J1. Test shall indicate open circuit. Remove plug and continuity shall be indicated.

(6) Repeat steps (2) through (5) for jacks 2 through 24, and 26 on strips A1 through A4.

(7) Connect one end of 25' cable CX4566/C to connector #1 and the other end to connector #2.

(8) Insert continuity tester into A3J1. Test shall indicate open circuit.

(9) Insert shorted plug into A1J1. Test shall indicate continuity.

(10) Insert dummy plug into A4J1. Test shall indicate continuity.

(11) Repeat steps 8 through 10 for jacks 2 through 24 on jack strips A1 through A4.

(12) Insert shorted plug into A1J26.

(13) Insert continuity tester into intercom jack in wiremold. Test shall indicate continuity.

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(14) Insert dummy plug into A4J26. Test shall indicate open circuit.

(15) Remove shorted plug from A1J26 and insert into A4J26. Test shall indicate continuity.

(16) Insert continuity tester into A1J25. Test shall indicate open circuit.

(17) Insert shorted plug into A4J25. Test shall indicate continuity.

(18) Repeat steps (2) through (17) for:

(a) Connectors 3 and 4 with jack strips A5 through A8.

(b) Connectors 13 and 14 with jack strips B1 through B4.

(c) Connectors 15 and 16 with jack strips B5 through B8.

(d) Connectors 25 and 26 with jack strips C1 through C4.

(e) Connectors 27 and 28 with jack strips C5 through C8.

(19) Connect one end of 25' cable CX4566/C to connector #3 and the other end to connector #6.

(20) Insert continuity tester into A10J1. Test shall indicate open circuit.

(21) Insert shorted plug into A13J1. Test shall indicate continuity.

(22) Insert dummy plug into A9J1. Test shall indicate open circuit. Remove dummy plug. Test shall indicate continuity.

(23) Insert dummy plug into A11J1. Test shall indicate open circuit. Remove dummy plug. Test shall indicate continuity.

(24) Insert dummy plug into A12J1. Test shall indicate open circuit. Remove dummy plug. Test shall indicate continuity.

(25) Insert dummy plug into A14J1. Test shall indicate open circuit. Remove dummy plug. Test shall indicate continuity. Remove shorted plug from A13J1.

(26) Insert continuity tester into A11J1. Test shall indicate open circuit.

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(27) Insert shorted plug into A14J1. Test shall indicate continuity.

(28) Repeat steps (20) through (27), for jacks 2 through 24 on jack strips A9 through A14 respectively.

(29) Insert shorted plug, into A11J26.

(30) Insert continuity tester into intercom jack in wiremold. Continuity shall be indicated.

(31) Insert dummy plug into A14J26. Test shall indicate open circuit.

(32) Remove shorted plug from A11J26.

(33) Insert shorted plug into A14J26. Test shall indicate continuity.

(34) Insert dummy plug into A11J26. Test shall indicate open circuit.

(35) Insert continuity tester into A11J25.

(36) Insert shorted plug into A14J25. Test shall indicate continuity.

(37) Transfer 25' cable to connectors 5 and 7.

(38) Insert continuity tester into A9J1. Test shall indicate open circuit.

(39) Insert shorted plug into A12J1. Continuity shall be indicated.

(40) Repeat steps (38) and (39) for jacks 2 through 24 on jack strips A9 and A12 respectively.

(41) Insert shorted plug into A9J26.

(42) Insert continuity tester into intercom jack in wiremold. Test shall indicate continuity.

(43) Insert dummy plug into A12J26. Test shall indicate open circuit.

(44) Remove shorted plug from A9J26.

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(45) Insert shorted plug into A12J26. Test shall indicate continuity.

(46) Insert dummy plug into A9J26. Test shall indicate open circuit.

(47) Insert continuity tester into A9J25.

(48) Insert shorted plug into A12J25. Test shall indicate continuity.

(49) Repeat steps (19) through (36), for connectors 10 and 12 with jack strips A15 through A20, substituting jack strips A15 for A9, A16 for A10, A17 for A11, A18 for A12, A19 for A13 and A20 for A14.

(50) Repeat steps (37) through (48) for connectors 9 and 11, with jack strips A15 and A18, substituting as in (49) above.

(51) Repeat steps (19) through (36) for connectors 18 and 20, with jack strips B9 through B14 respectively, substituting jack strips B9 for A9, B10 for A10, B11 for A11, B12 for A12, B13 for A13 and B14 for A14.

(52) Repeat steps (37) through (48) for connectors 17 and 19, with jack strips B9 and B12, substituting as in (51) above.

(53) Repeat steps (19) through (36) for connectors 30 and 32, with jack strips C9 through C14, substituting jack strips C9 for A9, C10 for A10, C11 for A11, C12 for A12, C13 for A13 and C14 for A14.

(54) Repeat steps (37) through (48) for connectors 29 and 31, with jack strips C9 and C12, substituting as in (53) above.

(55) Repeat steps (19) through (36) for connectors 34 and 36 with jack strips C15 through C20 respectively, substituting jack strips C15 for A9, C16 for A10, C17 for A11, C18 for A12, C19 for A13 and C20 for A14.

(56) Repeat steps (37) through (48) for connectors 33 and 35, with jack strips C15 and C18, substituting as in (55) above.

(57) Transfer CX4566/C cable to connectors 1 and 22.

(58) Insert continuity tester plug into B16J1.

(59) Insert shorted plug into A1J1. Test shall indicate continuity.

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(60) Insert dummy plug into B15J1. Test shall indicate open circuit.

(61) Remove dummy plug from B15J1. Test shall indicate continuity.

(62) Insert dummy plug into B17J1. Test shall indicate open circuit.

(63) Remove dummy plug from B17J1. Test shall indicate continuity.

(64) Repeat steps (58) through (63) for jacks 2 through 24 on strips A1, B15, and B17.

(65) Insert shorted plug into A1J26.

(66) Insert continuity tester plug into intercom jack in wiremold. Test shall indicate continuity.

(67) Insert dummy plug into B17J26. Test shall indicate open circuit.

(68) Remove dummy plug from B17J26. Test shall indicate continuity.

(69) Insert shorted plug into A1J25.

(70) Insert continuity tester plug into B17J25. Test shall indicate continuity.

(71) Remove shorted plug from A1J25. Test shall indicate open circuit.

(72) Transfer CX4566/C cable to connectors 1 and 21.

(73) Insert shorted plug into B16J1.

(74) Insert continuity tester into A2J1. Test shall indicate continuity.

(75) Repeat steps (73) and (74) for jacks 2 through 24 on strips A2 and B16.

(76) Insert shorted plug into A1J26.

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(77) Insert continuity tester plug into intercom jack in wiremold. Test shall indicate continuity.

(78) Insert dummy plug into B15J26. Test shall indicate open circuit.

(79) Remove dummy plug from B15J26. Test shall indicate continuity.

(80) Insert continuity tester into A1J25.

(81) Insert shorted plug into B15J25. Test shall indicate continuity.

(82) Remove shorted plug from B15J25. Test shall indicate open circuit.

(83) Insert continuity tester into A21J1. Test shall indicate open circuit.

(84) Insert shorted plug into A22J1. Test shall indicate continuity.

(85) Repeat steps (83) and (84) for A22J2 and A21J2.

(86) Insert continuity tester into A21J3. Test shall indicate open circuit.

(87) Insert shorted plug into A21J4. Test shall indicate continuity.

(88) Remove shorted plug from A21J4. Test shall indicate open circuit.

(89) Insert shorted plug into A21J5. Test shall indicate continuity.

(90) Repeat steps (86) through (89) for jack groups:

(a) A21J6 to A21J8

(b) A21J9 to A21J11

(c) A21J12 to A21J14

(d) A21J15 to A21J17

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(e) A21J18 to A21J20

(f) A21J21 to A21J23

(g) A21J24 to A21J26

(h) C21J1 to C21J3

(i) C21J4 to C21J6

(j) C21J7 to C21J9

(k) C21J10 to C21J12

(l) C21J13 to C21J15

(m) C21J16 to C21J18

(n) C21J19 to C21J21

(o) C21J22 to C21J24

(91) Insert resistance phone plug into A22J3. Place ohmmeter across tinned leads. Resistance shall read 35 ohm ± 2 ohms.

(92) Insert resistance phone plug into A22J4. Resistance shall read 332 ± 3 ohms.

(93) Insert resistance phone plug into A22J5. Resistance shall read 332 ± 3 ohms.

(94) Repeat steps (91) through (93) for the following groups of jacks:

(a) A22J6 to A22J8

(b) A22J9 to A22J11

(c) A22J12 to A22J14

(d) A22J15 to A22J17

(e) A22J18 to A22J20

(f) A22J21 to A22J23

(g) A22J24 to A22J26

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(h) B22J1 to B22J3

(i) B22J4 to B22J6

(j) B22J7 to B22J9

(k) B22J10 to B22J12

(l) B22J13 to B22J15

(m) B22J16 to B22J18

(n) B22J19 to B22J21

(o) B22J22 to B22J24

(95) Insert continuity tester into B23J1. Test shall indicate open circuit.

(96) Insert shorting plug into B23J2. Test shall indicate open circuit.

(97) Insert shorting plug into B23J5. Test shall indicate continuity.

(98) Remove shorting plug from B23J2 and insert reverse shorting plug. Test shall indicate open circuit. Replace shorting plug.

(99) Remove shorting plug from B23J5 and insert reverse shorting plug. Test shall indicate open circuit.

(100) Repeat steps (95) to (99) for jacks B23J22, B23J23, and B23J26.

(101) Insert continuity tester into B20J1. Test shall indicate continuity.

(102) Insert shorting plug into B20J2. Test shall indicate open circuit.

(103) Remove shorting plug from B20J2. Test shall indicate continuity.

(104) Insert reverse shorting plug into B20J2. Test shall indicate continuity. Remove plug.

(105) Insert shorting plug in B20J3. Test shall indicate open circuit.

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(106) Remove shorting plug from B20J3. Test shall indicate continuity.

(107) Insert reverse shorting plug into B20J3. Test shall indicate continuity.

(108) Repeat steps (101) to (107) for the following groups of jacks:

(a) B20J4 to B20J6

(b) B20J7 to B20J9

(c) B20J10 to B20J12

(d) B20J13 to B20J15

(e) B20J16 to B20J18

(f) B20J19 to B20J21

(g) B20J22 to B20J24

(h) B21J1 to B21J3

(i) B21J4 to B21J6

(j) B21J7 to B21J9

(k) B21J10 to B21J12

(l) B21J13 to B21J15

(m) B21J16 to B21J18

(n) B21J19 to B21J21

(o) B21J21 to B21J24

(109) Insert continuity tester in B20J25. Continuity shall be indicated.

(110) Repeat step (109) for B20J26.

(111) Transfer CX4566/C cable to connectors 23 and 24.

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(112) Insert continuity tester into B18J1 Test shall indicate open circuit.

(113) Insert shorted plug into B19J1. Test shall indicate continuity.

(114) Repeat steps 112 and 113 for jacks B18J2 through B18J25 and B19J1 through B19J25.

(115) Insert continuity tester into intercom jack in wiremold. Test shall indicate open circuit.

(116) Insert shorted plug into B18J26. Test shall indicate continuity.

(117) Insert shorted plug into B19J26. Test shall indicate open circuit.

(118) Remove shorted plug from B18J26. Test shall indicate continuity.

(119) Remove shorted plug from B19J26. Test shall indicate open circuit.

(120) Insert dummy plug into C21J26. Test shall indicate open circuit.

(121) Remove dummy plug from C21J26. Test shall indicate continuity.

(122) Remove shorting plug from A1J26. Test shall indicate open circuit.

(123) Short out intercom binding posts in power entrance box with diode, conducting from B to A. Test shall indicate continuity.

(124) Insert continuity tester into B20J25. Test shall indicate continuity.

(125) Insert continuity tester into B20J26. Test shall indicate continuity.

(126) Insert shorting plug into B23J7.

(127) Insert continuity tester into B21J26. Test shall indicate continuity.

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(128) Insert dummy plug into B21J25. Test shall indicate open circuit.

(129) Remove continuity tester from B21J26.

(130) Insert shorting plug into B23J7.

(131) Connect volt-ohm-meter (ME-77( )/U) to proper terminals. Set resistance scale to RX1.

(132) Insert shorting plug into B23J7. Ohm-meter shall indicate low resistance.

(133) Set voltage scale of meter to 30 volts.

(134) Insert continuity tester into B23J6. Voltmeter shall read voltage of continuity tester power supply.

(135) Insert continuity tester into phone jack in wiremold. Test shall indicate open circuit.

(136) Repeat step (123) for phone binding posts in power entrance box. Test shall indicate continuity.

(137) Remove jumper. Test shall indicate open circuit.

(138) Insert shorted plug into B22J25. Test shall indicate continuity.

(139) Remove shorted plug from B22J25. Test shall indicate open circuit.

(140) Insert shorted plug into B22J26. Test shall indicate continuity.

(141) Insert continuity tester into B23J8.

(142) Short out binding posts 1A and 1B with diode, conducting from 1A to 1B. Test shall indicate continuity.

(143) Repeat steps (135) through (137) for jacks B23J9 through B23J19, and binding posts 2A to 12A and 2B to 12B in sequence.

(144) Insert continuity tester into B23J20. Test shall indicate open circuit.

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(145) Connect shorting diode to TA248/TT leads, with diode conducting from wht-red-brn to wht-gy-brn. Test shall indicate continuity.

(146) Connect continuity tester across binding posts 13A and 13B. Test shall indicate open circuit.

(147) Short together wires of line 1 at SB-22 position. Test shall indicate continuity.

(148) Repeat steps (146) and (143) for binding posts 14A to 22A and 14B to 22B, with lines 2 to 10 at SB-22 position.

(149) Insert continuity tester into C19J25. Test shall indicate open circuit.

(150) Connect jumper across binding posts 23A and 23B. Test shall indicate continuity.

(151) Remove jumper from binding posts, and place across conductors of line 11 at SB-22 position. Test shall indicate continuity.

(152) Repeat steps (149) to (151) for C19J26, binding posts 24A and 24B and conductors of line 12.

4.6.2.3 Patch cords.- Using the continuity tester, test between the tips of each plug on each patch cord, between one sleeve and one tip of each patch cord. Continuity shall be indicated in each of the first two test, and discontinuity shall be indicated in the third test. (3.5.2.2)

4.6.2.4 Dielectric breakdown and insulation resistance, signal wiring.- The dielectric breakdown and insulation resistance of the signal wiring shall be tested for conformance with 3.5.2.3 (with either manual or automatic testing equipment) as described below:

a. Remove all telephone, teletype, patch cord, and intercom plugs from jacks in wireduct and patch panel. Check that lines are disconnected from SB-22 and TH-22's, and are not shorted to each other, the shelter, or any other metallic object.

b. Connect fixture No. 1 to connector No. 1.

c. Apply instantaneously a potential of 500 VDC between the fixture and the shelter ground, and hold this potential for at least 10 seconds.

d. Immediately after performing step c above, and with the full potential still applied, measure the insulation resistance. The resistance shall not be less than 100 megohms.

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e. Repeat steps b through d for connectors 2 through 36.

f. Insert the plugs of fixture No. 3 into the corresponding jacks on jack strip A21.

g. Repeat steps c and d.

h. Repeat steps f and g for jack strips A22, B20 through B23, and C21.

i. Connect fixture No. 5 to binding posts 13A to 22A, and 13B to 22B.

j. Repeat steps c and d for fixture No. 5.

k. Insert dummy pins in all jacks on strip A4.

l. Connect fixture No. 1 to connector No. 1.

m. Remove the jumper between binding posts 1L and 2L in fixture No. 1 junction box. (Hereafter the binding posts in the junction boxes will have two identifying marks. The first mark will indicate the number marked in the fixture box, and the second will indicate to the left (L) to right (R) of the number marked in the fixture box).

n. Repeat steps c and d, with the potential applied between 1L and 14R.

o. Shift the jumper from between 2L and 3L to between 1L and 2L.

p. Repeat step n.

q. Repeat steps o and p moving the jumper from between 3L and 4L to between A2L and A3L etc. in sequence from 4L to 13L, 13R to 1R, 14L to 26L and 26R to 14R.

r. Repeat steps k through q moving fixture No. 1 and the dummy plugs to the following connectors and jack strips:

(1) Connector 2 and jack strip A1.

(2) Connector 3 and jack strip A8.

(3) Connector 4 and jack strip A5.

(4) Connector 5 and jack strip A11.

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(5) Connector 6 and jack strip A9.

(6) Connector 7 and jack strip A14.

(7) Connector 8 and jack strip A12.

(8) Connector 9 and jack strip A17.

(9) Connector 10 and jack strip A15.

(10) Connector 11 and jack strip A20.

(11) Connector 12 and jack strip A18.

(12) Connector 13 and jack strip B4.

(13) Connector 14 and jack strip B1.

(14) Connector 15 and jack strip B8.

(15) Connector 16 and jack strip B5.

(16) Connector 17 and jack strip B11.

(17) Connector 18 and jack strip B9.

(18) Connector 19 and jack strip B14.

(19) Connector 20 and jack strip B12.

(20) Connector 21 and jack strip B17.

(21) Connector 22 and jack strip B15.

(22) Connector 23 (omit all dummy plugs).

(23) Connector 24 (omit all dummy plugs).

(24) Connector 25 and jack strip C4.

(25) Connector 26 and jack strip C1.

(26) Connector 27 and jack strip C8.

(27) Connector 28 and jack strip C5.

(28) Connector 29 and jack strip C11.

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(29) Connector 30 and jack strip C9.

(30) Connector 31 and jack strip C14.

(31) Connector 32 and jack strip C12.

(32) Connector 33 and jack strip C17.

(33) Connector 34 and jack strip C15.

(34) Connector 35 and jack strip C20.

(35) Connector 36 and jack strip C18.

s. Insert the plugs of the fixture No. 3 into the following jacks: A21J1, A21J2, B23J8, A21J6, A21J9, A21J12, A21J15, A21J18, A21J21, A21J24, C21J1, C21J4, C21J7, C21J10, C21J13, C21J16, C21J19, C21J22, C19J25, C19J26. NOTE: The jacks referenced in step s will hereafter be referred to as "Group 1".

t. Repeat steps m through q.

u. Repeat steps s and t, for the following jacks: A21J3, B23J9, B23J10, B23J11, B23J12, B23J13, B23J14, B23J15, B23J16, B23J17, B23J18, B23J20, C21J26. NOTE: The jacks referenced in step u will hereafter be referred to as "Group 2".

v. Insert dummy plugs into the following jacks: B20J2, B20J5, B20J8, B20J11, B20J14, B20J17, B20J20, B20J23.

w. Repeat steps s and t, for the following jacks: B20J1, B20J4, B20J7, B20J10, B20J13, B20J16, B20J19, B20J22, B22J25, B23J1, B23J22.

x. Remove dummy plugs from the jacks of step v.

y. Insert dummy plugs into the following jacks: B20J3, B20J6, B20J9, B20J12, B20J15, B20J18, B20J21, B20J24.

z. Repeat steps s and t, for the following jacks: B20J2, B20J5, B20J8, B20J11, B20J14, B20J17, B20J20, B20J23, B23J2, B23J23, C21J1, and the intercom jack in wiremold. NOTE: The jacks referenced in steps w and z will hereafter be referred to as "Group 3".

aa. Connect the bare leads of fixture No. 5 to binding posts 11A to 22A, and 11B to 22B. (The binding posts referenced in step aa will hereafter be referred to as "Group 4").

bb. Repeat step t.

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cc. Place shorted connectors on each signal connector of shelter facility.

dd. Insert the plugs of fixture No. 3 into the jacks of "Group 1".

ee. Insert the plugs of fixture No. 4 into the jacks of "Group 2".

ff. Connect the leads of fixture No. 5 to the binding posts of "Group 4".

gg. Insert the plugs of fixture No. 7 into the jacks of "Group 3".

hh. Run patch cables between the following jacks:

(1) A3J1 to A6J1

(2) A7J1 to A10J1

(3) A11J1 to A13J1

(4) A14J1 to A16J1

(5) A17J1 to A19J1

(6) A20J1 to B2J1

(7) B3J1 to B6J1

(8) B7J1 to B10J1

(9) B11J1 to B13J1

(10) B14J1 to B16J1

(11) B17J1 to B19J1

(12) B19J2 to B20J1

(13) B20J2 to C2J1

(14) C3J1 to C6J1

(15) C7J1 to C10J1

(16) C11J1 to C13J1

(17) C14J1 to C16J1

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(18) C17J1 to C19J1

(19) C20J1 to A22J1

(20) A22J2 to A21J4

(21) A21J5 to C21J2

ii. Terminate the leads of a "hi-pot" tester in two phone jacks, with the tip and ring of one jack bussed together, and tied to the positive lead of the "hi-pot" tester, and a similar hookup with the negative lead, and the other jack.

jj. Insert one jack of the hi-pot tester into A2J1, and the other jack into B23J19.

kk. Insert dummy plugs into all intercom jacks with the exception of the jack in the wiremold. NOTE: These dummy plugs will remain in place for the duration of the test.

ll. Repeat steps c and d.

mm. Remove the patch cable plug from A21J5 and insert into B23J18.

nn. Repeat steps c and d.

oo. Remove the patch cable plug from A22J2, and insert into C21J3.

pp. Repeat steps c and d.

qq. Remove the patch cable plug from C20J1, and insert into A21J5.

rr. Repeat steps c and d.

ss. Connect patch cord between C20J1 and A22J2.

tt. Insert dummy plugs into all empty jacks on strip C20.

uu. Repeat steps c and d.

vv. Remove the dummy plugs of step tt.

ww. Repeat steps tt through vv for the following jack strips:

(1) C18

(2) C17

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(3) C15

(4) C14

(5) C12

(6) C11

(7) C9

(8) C8

(9) C5

(10) C4

(11) C1

(12) B17

(13) B15

(14) B14

(15) B12

(16) B11

(17) B9

(18) B8

(19) B5

(20) B4

(21) B1

(22) A20

(23) A18

(24) A17

(25) A15

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(26) A14

(27) A12

(28) A11

(29) A9

(30) A8

(31) A5

(32) A4

(33) A1

xx. Remove the patch cord connecting C17J1 and C19J1.

yy. Repeat steps c and d.

zz. Replace the patch cord of step xx.

aaa. Repeat steps xx through zz for the patch cords between the following jacks:

(1) C14J1 and C16J1

(2) C11J1 and C13J1

(3) C7J1 and C10J1

(4) C3J1 and C6J1

(5) B20J2 and C2J1

(6) B19J2 and B20J1

(7) B17J1 and B19J1

(8) B14J1 and B16J1

(9) B11J1 and B13J1

(10) B7J1 and B10J1

(11) B3J1 and B6J1

(12) A20J1 and B2J1

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(13) A17J1 and A19J1

(14) A14J1 and A16J1

(15) A11J1 and A13J1

(16) A7J1 and A10J1

(17) A3J1 and A6J1

4.7 Military truck transport.- The shelter facility with operational equipment (3.3) installed shall be transported on a Government furnished M-35 truck over the specially prepared course described in paragraph 4.7.1 for 10 laps at 5 mph, 10 laps at 10 mph, 10 laps at 15 mph and 10 laps at 20 mph. A lap is defined as traversing the course in one direction. The shelter facility shall meet the requirements of 3.6 after completion of the above test.

4.7.1 Military truck transport course.- The course for the military truck transport test shall have twelve 4-inch by 8-inch timbers placed 25 feet apart on the 8-inch face and with the 4-inch face fully above the ground. The edges shall have a 1-inch by 1-inch bevel. The sixth and twelfth timbers shall be placed 45 degrees to the direction of travel, and all other timbers shall be placed perpendicular to the direction of travel. The timbers shall be long enough to span the vehicle passing over them and shall be anchored securely by any convenient means. Ground on which timbers are placed shall be hard and level.

4.8 Rail transport, impact test.- The shelter facility with operational equipment (3.3) installed shall be loaded on a railroad flatcar (test car) in accordance with Figure 1. The test shall be conducted on a flat stretch of track. A loaded car totaling 165,000 pounds (impact car) and traveling at 9 mph (or equivalent impact energy conditions approved by the Government) shall be impacted against the test car coupled to two other empty cars. These cars shall be stationary with all brakes off. Four impacts shall be performed, two impacts into each end of the test car. The cabling and blocking holding the shelter facility in place shall be tight at the start of each impact run. If the cable or blocking is torn loose or if the load on the impact car shifts, the run shall be repeated. A damage survey shall be made after each impact and the shelter facility shall meet the requirements of 3.7.

4.9 Drop tests.-

4.9.1 Flat drop.- The shelter facility with operational equipment (3.3) installed shall be lifted and allowed to fall freely a distance of 18 inches with the skids impacting onto a hard concrete surface. This test shall be performed once. Upon completion of this drop, the shelter facility shall meet the requirements of 3.8.

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ITEM	REQ.	DESCRIPTION
1	2	LUMBER 4"x5"x18'
2	1	LUMBER 6"x6" TO SUIT
3	2	LUMBER 6"x6" TO SUIT
4	2	LUMBER 8"x8" TO SUIT
5	2	LUMBER 4"x4"x13'
6	10	LUMBER 2"x4"x14"
7	2	LUMBER 6"x6" TO SUIT
8	1	LUMBER 6"x6" TO SUIT
9	2	LUMBER TO SUIT
10	16	WIRE #8 SOLID, HARD, 2250 LB TEST
11	4	FLEX. METAL SHEETING

Wire tiedown same for all points of contact with stake pockets

NOTES:

a. The blocking lumber shall be notched to accomodate towing eyes, bolts and rivet heads.

b. The blocking lumber shall be rigidly fastened to the floor of the flatcar.

FIGURE 1

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4.9.2 Rotational drop.- The shelter facility with operational equipment (3.3) installed shall have one edge placed on a nominal 4 inch high board. The opposite edge of the shelter facility shall be raised 18 inches and allowed to fall freely onto a hard concrete surface. This drop test shall be performed on each bottom edge of the shelter facility. Upon completion of these drops, the shelter facility shall meet the requirements of 3.8. In each drop test, the 4 inch high member shall be placed under the skids and the eighteen inch measurement shall be taken from the bottom of the skids.

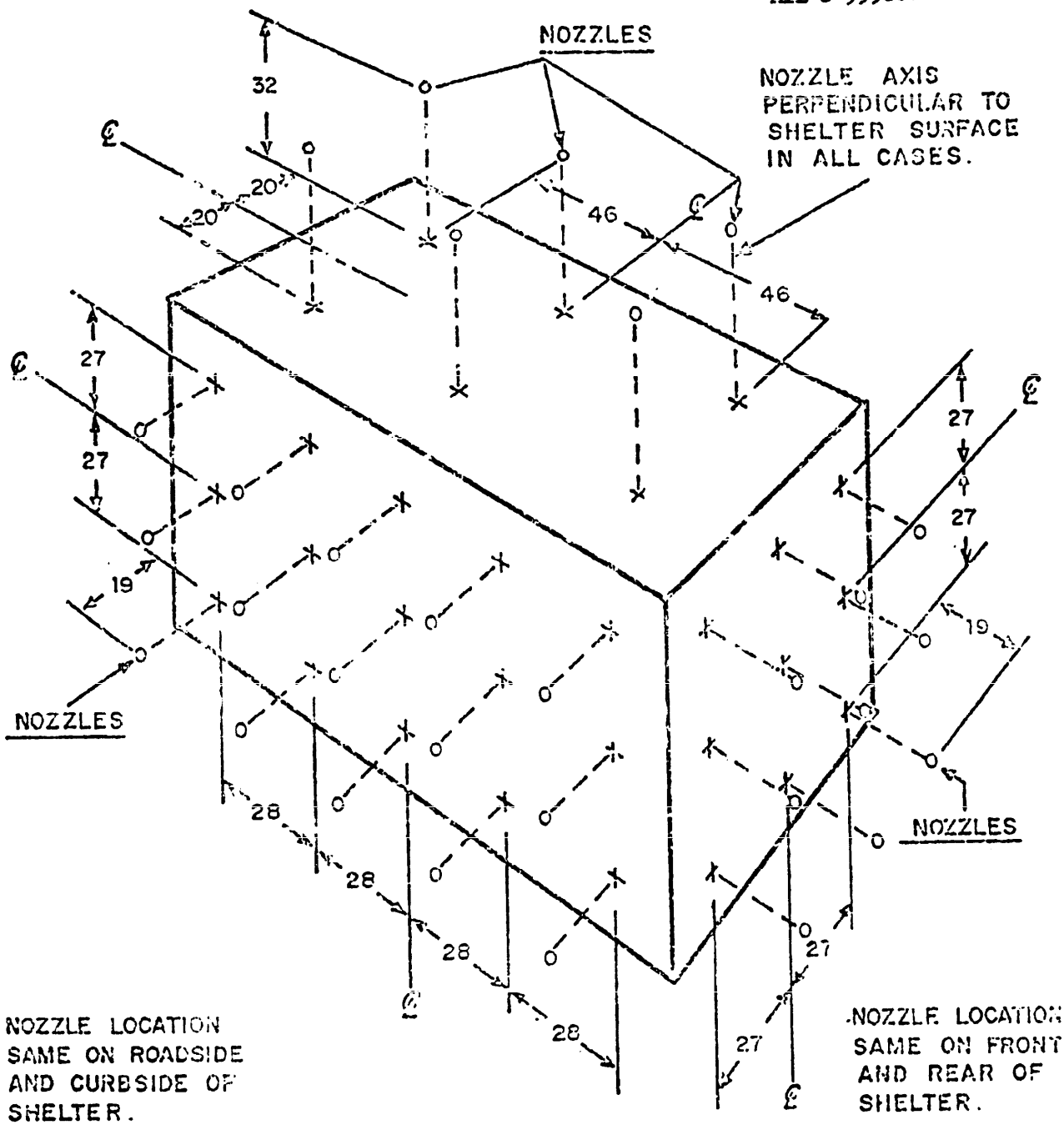
4.10 Watertightness test.- The shelter facility without the use of any additional caulking, sealing, taping, etc, shall be subjected to the following test: Each of the five exposed surfaces of the shelter, with covers opened, shall be sprayed with water from nozzles, model G29SQ (or GG29SQ) as made by Spraying Systems Co., Bellwood, Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using fifteen nozzles, and the roof using six nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches and the row of nozzles in line with the air conditioner where the 19 inches shall be measured from the extreme vertical surface of the air conditioner. (See Figure 2 for nozzle location). Each panel shall be sprayed for a period of forty minutes. (If the watertight test is to be followed by the fording test of 4.11, the bottom row of nozzles, with the exception of the nozzles facing the door-end panel, can be eliminated.) Upon completion of the above test, the shelter facility shall meet the requirements of 3.9. To determine that no water has entered space between the skins, holes shall be made at points to be determined by the Government at time of testing. After this determination, the holes shall be sealed with rivets and sealer.

4.11 Fording test.- The shelter facility, prepared for transportation on a military cargo truck and without the use of any additional sealer, caulking, taping, fording kit, etc, shall be immersed in water to a depth of twenty-one inches, measured from the bottom of the shelter skids, for a period of two hours. The shelter shall be restrained by external tiedowns. Upon completion of this test, the shelter facility shall meet the requirements of 3.10. To determine that no water has entered space between the skins, holes shall be made at points to be determined by the Government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

4.12 Testing of solderless terminal-lug connections.- Connections made with solderless terminal-lugs shall be tested to meet the requirements of 3.19.5.2.

4.13 Insert test.- Two percent (2%) of the total number of any size insert, but not less than five inserts, shall be randomly selected for test in each sample facility. Each selected inserts shall have a torque load, in accordance with Table V applied to the bolt. NOTE: Bolt shall hold a stationary plate against head of insert. After test, the requirements of 3.11 shall be met. Failure of any insert shall be cause for rejection of sample facility.

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Note :

GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES.

FIGURE 2

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Table V - Production Torque Load

Insert size	Torque (in lb)
5/16 - 18	100
1/4 - 20	60
10 - 32	30
8 - 32	20

4.14 Sealer shear tests.- A tensile pull test shall be performed on sample lap joints consisting of two (2) one-inch wide strips of aluminum (0.062 inch thick) overlapped one-half inch, giving contact area of one-half square inch and bonded with sealer material over no more than the specified one-half square inch area. The samples shall be tested over the conditions of 3.20.1 through 3.20.3. Five specimens shall be used for each test condition and tests shall be to destruction with rate of application of load constant at one inch per minute maximum.

4.15 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.21. When a dimension is not within specified or design limits, it shall be considered a defect.

a. Location and size of mounting inserts.

b. Size and mounting dimensions of brackets.

c. Size and antenna pass-through plugs.

d. Size and location of radio shelf mounting nuts.

e. Size and mounting dimensions of blowers, and heaters.

4.16 Visual and mechanical inspection.- Equipment shall be examined for the defects listed in Standard MIL-STD-252 and Table VI.

Table VI - Classification of Visual and Mechanical Defects

Classification	Defects
Major	1. Mounting insert missing or mis-located.

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Major	2.	Sealer improperly applied (holes, separations, or lack of adhesion).
	3.	Welds - cracked or porous.
	4.	Parts, brackets, or equipment loose or missing.
	5.	Drawers, doors, etc, inoperative.
	6.	Blower inoperative or rotating in wrong direction.
	7.	Locking or holding devices inoperative.
	8.	Lock washer or lock nut missing.
	9.	Burrs and sharp edges in wireduct not removed.
	10.	Incorrect or missing labels on binding parts and jacks.
Minor	1.	Screw loose.
	2.	Drawers, doors, etc, difficult to operate.
	3.	Locking or holding devices difficult to operate.
	4.	Finish - abrasions or scratches.
	5.	Burrs and sharp edges on brackets not removed.

4.17 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-55505.

4.18 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract (See 6.1), the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling: Paragraph 4.15 Visual and Mechanical.

5. PREPARATION FOR DELIVERY.-

5.1 Preservation and packaging.- Preservation, packaging and packing of Shelter Facility S-404/MSC shall be in accordance with the applicable requirements of Specification MIL-P-55505.

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6. NOTES.-

6.1 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment.

b. Level of packaging and level of packing required for shipment (level A, level B, or level C).

c. The specific paragraphs of section 5 which are applicable to particular procurement.

d. Preproduction pack(s) as follows:

(1) Makeup of pack(s).

(2) Number of each kind of pack to be submitted.

(3) Inspection to be performed thereon - including rough handling test, which will not be performed as procurement inspection.

(4) Five copies of process sheet, using format shown on Drawing SC-C-33073 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. This data should be submitted to the contracting officer with the preproduction pack.

(5) A packaging technician from the procuring agency will evaluate the preproduction pack(s).

e. Marking and shipping of samples.

f. Place of final inspection.

g. List of each individual equipment item or equivalent dummy load, with details, to be used in testing the shelter facility (3.3).

h. Submission of the statement of treatment referenced in 3.17.4, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.2 Definitions.-

6.2.1 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

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6.2.2 Examination.- Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.2.3 Testing.- Testing consists of determinations of compliance, using technical means.

6.4 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: S-404X/MSC. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.5 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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